

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A

(Objective Type Questions)

1. Choose the correct answer in each of the
following : 1×5 = 5

(a) HTTP stands for :

- (i) ☒ Hyper Text Transfer Protocol
- (ii) Hyper Transfer Text Protocol
- (iii) Hyper Transmission Text Protocol
- (iv) None of these

- (b) FDM stands for :
- (i) Frequency Division Multiplexing
 - (ii) Frequency Division Multiple
 - (iii) Frequency Division Motion
 - (iv) None of these
- (c) What is Number of Layers in the OSI Model ?
- (i) 6 Layers (ii) 5 Layers
 - (iii) 9 Layers (iv) 7 Layers
- (d) Identify the Layer which provides services to the users :
- (i) Application Layer
 - (ii) Session Layer
 - (iii) Presentation Layer
 - (iv) Physical Layer
- (e) POP stands for :
- (i) Pre-Office Protocol
 - (ii) Post Office Protocol
 - (iii) Protocol Of Post
 - (iv) None of these

2. Fill in the blanks :

1×5 = 5

- (a) _____ is set of rules that consists the data communication.

- (b) _____ Layer takes raw transmission facility and transform it in to line.
- (c) _____ is the basic protocol of the Internet.
- (d) CSMA/CD stands for _____.
- (e) Intranet is also called _____.

Group – B

(Short-answer Type Questions)

Answer any **four** questions of the following :

$5 \times 4 = 20$

3. (a) What are LAN Topologies ? Describe the types of LAN topologies with Diagram.
- (b) Describe the different types of protocol with Diagram.
- (c) Explain the Analog Signal and Digital Signal in Data Communication.
- (d) Explain the Pulse Code Modulation (PCM).
- (e) Explain the Serial transmission in Data Communication.
- (f) Explain the following (any **two**) :
- (i) Stop and Wait ARQ
 - (ii) Router

(iii) Repeater

(iv) Gateway

Group – C

(Long-answer Type Questions)

4. Answer any **four** questions of the following :

10×4 = 40

- (a) What is OSI model ? Describe the different layer of the OSI Model with Diagram.
- (b) Differentiate between Circuit Switching and Packet Switching.
- (c) Describe the Handshaking Method of the transport Layer.
- (d) Explain the Error correction and detection of the Data Link Layer.
- (e) Explain the multiplexing technique.
- (f) Write short notes of any **two** of the following :
 - (i) HTTP
 - (ii) TCP/IP
 - (iii) TDM
 - (iv) Transmission Media

40/0